

GitSkills: dataset profiling

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Summary

GitSkills at full scale shows concentrated reuse and only a weak link between bundled scripts and how widely a skill spreads. For Assignment 1, adapted to the GitSkills dataset of the MSR 2027 Mining Challenge and run on all 3.8M files.

- 3,797,117 SKILL.md files from 282,200 public GitHub repos (July 2026), grouped into 1,877,981 distinct contents. 50.5% of files are exact copies of another file.
- Reuse is concentrated: 79.3% of distinct skills appear once, 312 repos (0.11%) hold 36.7% of all skill files, and the top 1% of contents account for 25.9% of all occurrences.
- 11.4% of skills bundle scripts (mostly Python, TypeScript, JavaScript and Shell) and 25.9% bundle reference files. Bundled files total 200.8 GB, and almost half are Markdown.
- Skills that bundle scripts are only weakly more likely to spread widely. Outside collection repos, 0.81% of script skills and 0.70% of other skills sit in 10 or more repos (odds ratio 1.15, 95% CI 1.08 to 1.23), and the size of the effect depends on how collection repos are defined.
- Collection repos carry most of the copying: about three quarters of the copies of widely spread skills sit in repos with 100 or more skill files. About a dozen repos look like wholesale clones of one security-skills collection, which is not yet verified.

1. Data and method

Data. GitSkills (Destefanis, Graziotin, Vaccargiu and Ortu, arXiv:2608.10906) is a July 2026 snapshot of public GitHub repositories and a lower bound on the population. A skill is a folder with a SKILL.md file of instructions for an AI coding agent, optionally with scripts and reference files. Skills have no registry or package manager, so they spread by copying folders.

Method. The Parquet mirror was queried with DuckDB in Google Colab, after the code was developed on the authors' 13,000-skill sample. Row counts match the paper exactly. Group comparisons report odds ratios with 95% Wald intervals.

Text analyses (URLs, language, vocabulary) use a deterministic 1/40 sample of distinct skills by content hash: 47,160 skills, with language detection on the first 20,000.

2. Schema

The data has four tables, and only one representative copy per distinct content carries text and folder details.

Table	Rows	One row is	Main columns
artifacts	3,797,117	a SKILL.md file occurrence	repo, path, location class, content hash, representative flag. For the representative of each content: text, front matter, body size, has_scripts, has_references. History columns for a subset.
artifact_siblings	7,264,865	a file or folder bundled with a representative skill	entry name, type, size, text (if small)
repos	282,200	a repository	owner, stars, forks, language, license, dates
mining_runs	7	a collection run	query, timestamps, result count

The 1,877,981 representatives carry text, front matter and folder information. Copies carry location and hash only. The representative is chosen with a preference for .claude/skills/, so the location of distinct skills is biased and the location mix in section 3 uses all occurrences. Folder information and commit history describe the representative's repository only.

Not in the data: a skill domain or category; which copy is the original and who copied from whom; whether an agent ever uses a skill; popularity per skill (stars belong to the repo); human or AI authorship; what bundled scripts do.

Easy to answer: counts, sizes, locations, bundling and exact copies. **Hard to answer:** causes, quality, real usage and near duplicates.

Data notes: 1,544 representatives have no folder listing. The rule behind `has_scripts` is not documented in the preprint. The flag is a subset of a file-extension definition: 11,507 skills (5.1% of extension-based script skills) have script-extension files but flag 0.

3. Size metrics

The 3.8M occurrences collapse to 1.88M distinct contents, and bundled files add 5.9M files totalling 200.8 GB.

Quantity	Value
File occurrences / distinct contents	3,797,117 / 1,877,981 (50.5% of occurrences are copies)
Repos / owners	282,200 / 195,841
Location, all occurrences	canonical (.claude/skills/) 373,652 (9.8%); skills-dir 2,102,053 (55.4%); other 1,321,412 (34.8%)
Valid YAML front matter, distinct skills	1,625,701 (86.6%)
Bundle scripts (<code>has_scripts</code>), of 1,876,437 with a folder listing	214,507 (11.4%). Nested skills 198,505 of 1,851,519 (10.7%); root-level SKILL.md 16,002 of 24,918 (64.2%)
Bundle reference files	485,979 (25.9%). Nested 25.0%; root-level 96.3%
Bundled files / folders	5,858,945 files (200.8 GB) / 1,405,920 folders. 3,497,752 files (59.7%) have stored text
Skills with commit history / author codes	458,548 / 95,881 (anonymised, includes bots)

Root-level skills (SKILL.md at the repo root) are 1.3% of skills, but their folder is probably the whole repository, which inflates their bundling numbers. Nested skills are used for the comparisons in section 6.

Both size variables have a heavy right tail, so medians are reported next to means.

Statistic	Skill body (characters)	Bundled file (bytes)
n	1,877,981	5,858,945

Statistic	Skill body (characters)	Bundled file (bytes)
mean	6,958	34,273
median	4,600	3,558
std	9,160	626,178
min	0	0
q25	2,129	1,221
q75	8,664	8,734
max	355,781	104,857,563
skew	7.55	75.3
kurtosis	132	8,190

4. Distributions

Reuse is concentrated: 79.3% of distinct contents occur once, and 312 repos hold 36.7% of all skill files.

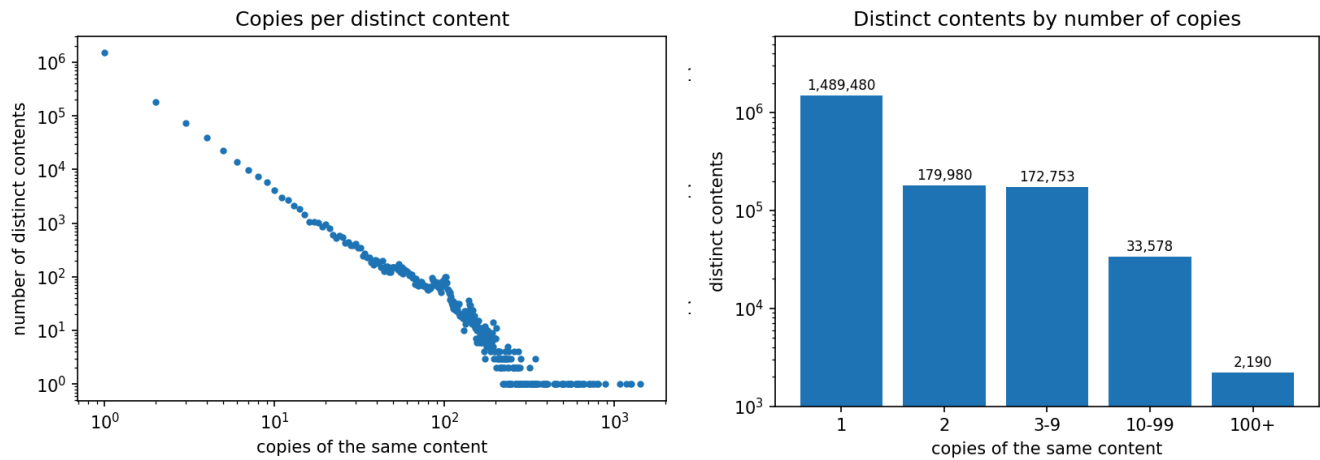


Figure 1. Copies per distinct content. 79.3% of contents occur once and 83.7% sit in one repo. The most copied content has 1,409 copies in 951 repos. The log-log plot is close to a straight line over about two decades, a heavy tail; no power law was fitted.

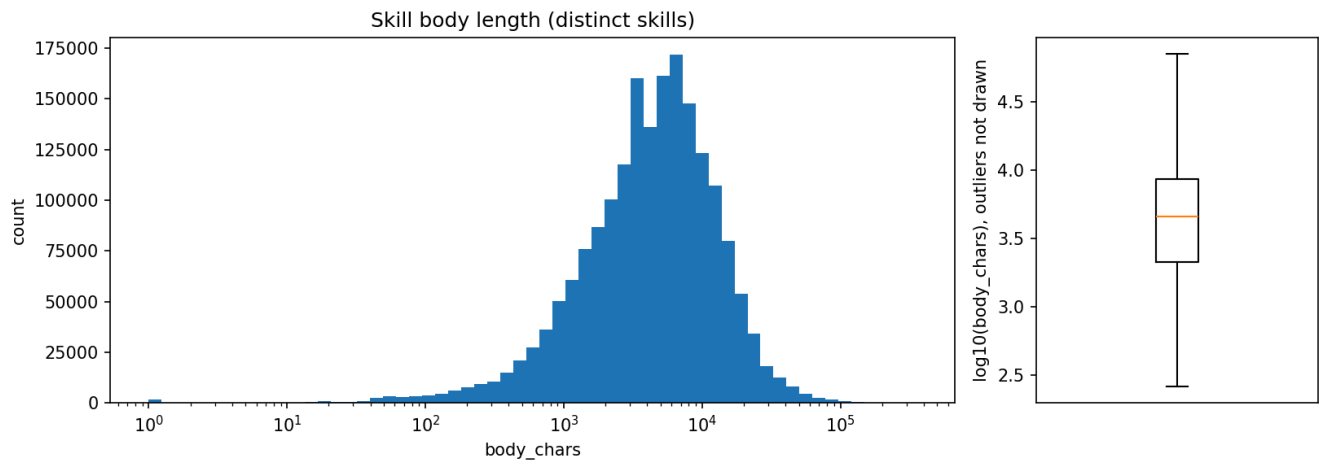


Figure 2. Skill body length, distinct skills (log scale). Median 4,600 characters, most between 1,000 and 10,000, with a thin tail of near-empty skills and a long tail up to 355,781.

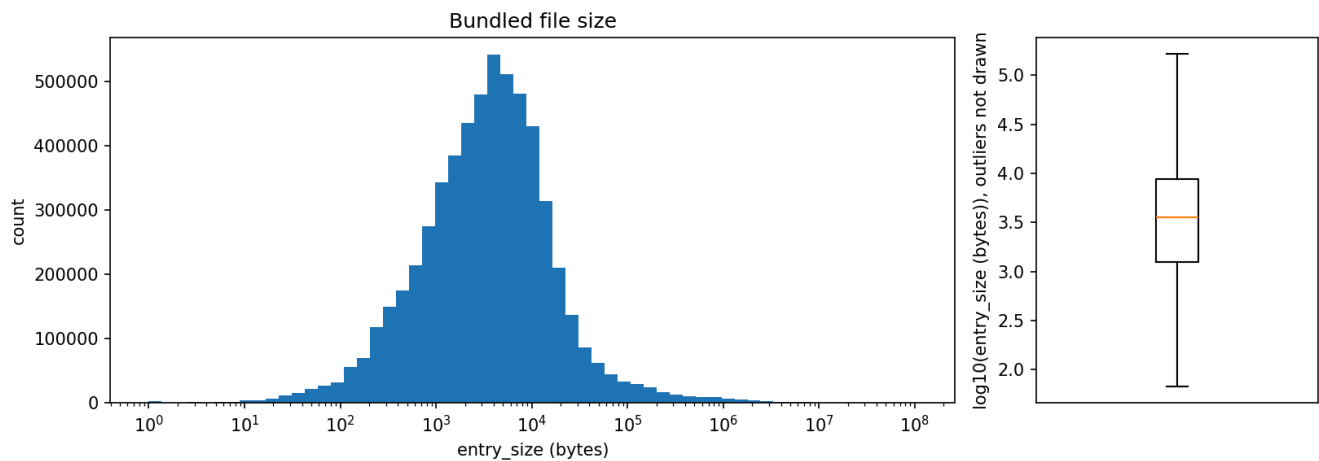


Figure 3. Bundled file size (log scale, zero-byte files omitted). Median 3.6 KB, mean 34 KB, maximum 100 MB (about the size cap). The 200.8 GB total is dominated by a few very large files. In the sample the largest were an installer, archives and model weights.

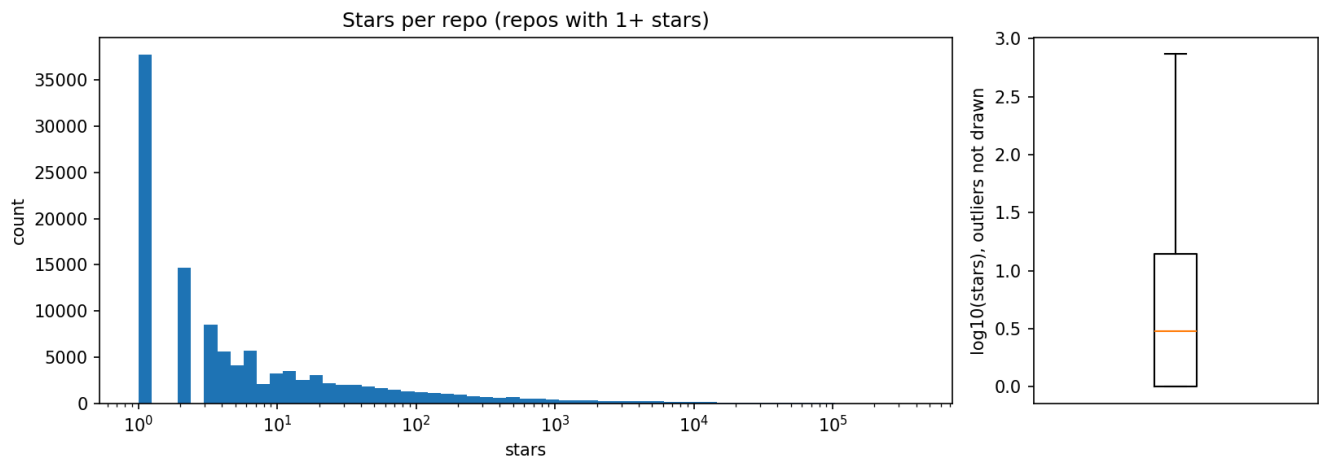


Figure 4. Stars per repo, repos with at least one star (log scale). 59.3% of repos have no stars, and the median among starred repos is 3.

Skill files per repo, all repos:

Skill files in repo	Repos	Share of all skill files
1	121,427	3.2%
2 to 5	86,086	6.9%
6 to 20	53,915	14.7%
21 to 100	17,396	18.2%
101 to 1,000	3,064	20.4%
1,001 or more	312	36.7%

5. Traceability

Bundled files are mostly Markdown, and 24.6% of skills link to a real external URL.

Bundled files. Of 5,858,945 bundled files, 3,497,752 (59.7%) have stored text. By extension: .md 2,804,074 (47.9%), .py 619,273, .json 371,240, .ts 234,527, .sh 141,203, .js 138,471, .yaml 137,330 and .png 102,304. Script-extension files total 1,380,580 in about 226,000 skills, and 85.8% of those skills use a single script language.

Script language	Files	Share
Python	619,273	44.9%
TypeScript	234,527	17.0%
JavaScript	208,517	15.1%
Shell	141,474	10.2%
Go	97,097	7.0%
Rust	27,181	2.0%
PowerShell	14,657	1.1%
Java	13,971	1.0%

Script language	Files	Share
PHP	13,230	1.0%

URLs (text sample, 47,160 skills). 24.6% of skills contain a real external URL (11,591), and there are 70,954 URLs in total. Placeholders such as example.com and localhost were removed with a simple filter. The most frequent domains, by skills mentioning them, are github.com (4,702), raw.githubusercontent.com (412), w3id.org (329), edamontology.org (324), learn.microsoft.com (222), arxiv.org (180), rube.app (174) and npmjs.com (148). The w3id.org and edamontology.org counts are almost equal, which suggests one copied template family.

Natural language (first 20,000 of the text sample, automatic detection). English is 90.0%, Chinese 1.6%, German 1.4%, Japanese 1.2%, Portuguese 1.0%, Norwegian 0.9%, Spanish 0.8%, Vietnamese 0.7% and Korean 0.6%. Detection is noisy on short or mixed text (Norwegian and Catalan are suspect) and has not been checked by hand.

Vocabulary (same sample). 1,169,098 unique tokens in 45.9M tokens and 8.8M lines, about 970 tokens per skill. A token is a lowercased word-character match, and vocabulary grows with sample size.

Over time. Only 83 of 451,038 first commits predate October 2025, when the format was introduced. New skills per month grow from 1,214 in October 2025 to about 85,000 in each of May and June 2026, and July 2026 is a partial month.

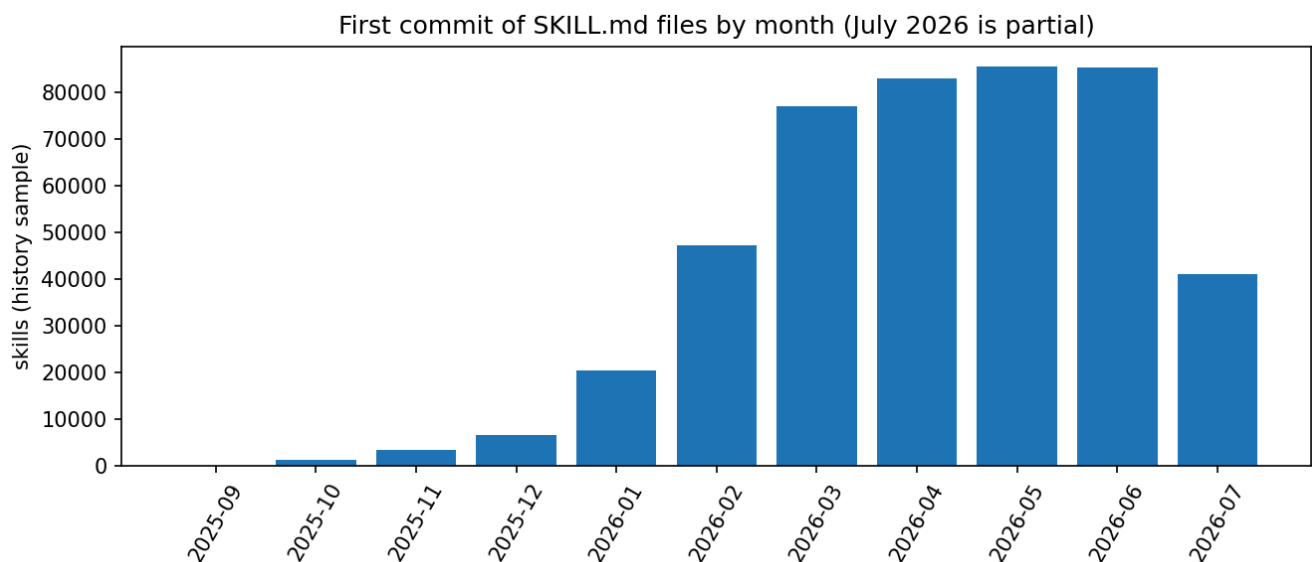


Figure 5. First commit of the SKILL.md file by month, for the representative copy of each skill with history, exact SKILL.md name only (n = 451,038). This subset is not a random sample.

6. Focused analysis: do skills that bundle scripts spread differently?

Script skills are only slightly more likely to spread widely, and the result depends on how collection repos are defined.

Definitions. Bundles scripts is the dataset's has_scripts flag, taken from the representative's folder. Spread is the number of distinct repos holding the identical content. A collection repo holds 100 or more skill files, widely spread means 10 or more repos, and nested excludes a root-level SKILL.md. The main comparison and its cutoffs were fixed before the full-data run. An odds ratio (OR) above 1 means script skills are more likely to spread.

Setting	n: no scripts / scripts	In 2+ repos	OR (95% CI)	In 10+ repos	OR (95% CI)
All repos, all skills (naive)	1,661,930 / 214,507	16.36% vs 15.78%	0.96 (0.95 to 0.97)	1.30% vs 1.99%	1.54 (1.49 to 1.59)
All repos, nested	1,653,014 / 198,505	16.44% vs 16.82%	1.03 (1.02 to 1.04)	1.31% vs 2.15%	1.66 (1.60 to 1.71)
MAIN: exclude repos with 100+ files, nested	997,335 / 126,568	9.16% vs 9.89%	1.09 (1.07 to 1.11)	0.70% vs 0.81%	1.15 (1.08 to 1.23)
Exclude repos with 20+ files	613,976 / 79,606	7.31% vs 7.93%	1.09 (1.06 to 1.12)	0.42% vs 0.46%	1.08 (0.97 to 1.20)
Exclude repos with 1000+ files	1,245,414 / 168,409	11.10% vs 12.97%	1.19 (1.18 to 1.21)	1.33% vs 1.86%	1.41 (1.35 to 1.46)
MAIN, exact SKILL.md basename only	984,758 / 124,052	9.17% vs 9.97%	1.10 (1.07 to 1.12)	0.71% vs 0.82%	1.16 (1.09 to 1.24)

p-values are not shown. With millions of skills they are always tiny, and skills cluster inside repos, so the intervals are too narrow. The size and direction of the odds ratios are what matter.

- **Main result (small).** Outside collection repos, script skills are slightly more likely to sit in 10+ repos (0.81% vs 0.70%, OR 1.15). The absolute difference is 0.11 percentage points.

- **Not stable.** The 10+ odds ratio is 1.41 when only the 312 largest repos are excluded, 1.15 at the main cutoff, and 1.08 (interval includes 1) when repos with 20+ files are excluded. The 2+ outcome keeps its direction outside collection repos (OR 1.09 to 1.19) but is close to 1 when collection repos are counted.
- **Collections dominate.** 75.1% of the copies of widely spread skills without scripts and 73.5% of those with scripts sit in repos with 100+ skill files. Collection repos, not scripts, are the main route by which skills spread.
- **Sample vs full data.** A first run on the 13,000-skill sample suggested an odds ratio near 2 for 10+ repos. The full data does not support that at the main setting, so the sample estimate should not be used.

Which skills are widely spread and bundle scripts? The most spread are webapp-testing (781 repos), several variants of pdf, pptx, xlsx and docx (about 370 to 660 repos each), systematic-debugging (656), mcp-builder (555), skill-creator (532 for one variant), slack-gif-creator (469) and algorithmic-art (458). Many look like Anthropic's public example skills and a few popular community packs. Their source repos have not been checked.

Where do they live? Two mirror or registry repos hold 1,952 and 1,251 of them (agent-skills-mirror and claude-skill-registry-data). Thirteen of the top 20 repos have only 664 to 952 skill files each yet hold 573 to 674 of these skills. They match the size of mukul975/Anthropic-Cybersecurity-Skills (790 files, 674 of them), which suggests one security-skills collection was cloned wholesale. This is not yet verified by comparing content sets and creation dates.

7. Limitations and next steps

The findings are associations from exact-copy matching, and several of them are exploratory.

- Copies are exact-hash matches only, so lightly edited copies are not counted. Folder information (has_scripts, siblings) comes from one representative copy per content, which prefers .claude/skills/.
- The has_scripts rule is undocumented. The comparison is an association, and domain and collection membership are confounders.
- Several cuts of the data were examined. Only the main comparison was planned in advance, and the others are exploratory.
- Language detection and the URL filter are heuristic and not hand-checked.

- The history subset is not random, and the dataset covers only what GitHub code search indexes (default branches, files under 384 KB, recently active repos).

Next steps. Verify the clone family (shared content, creation dates, stars). Read what widely spread scripts do, such as network calls and shell execution. Define a skill-domain taxonomy to study a specific context instead of the whole dataset. Ask the dataset authors for the `has_scripts` rule.

Sources

- [GitSkills: A Dataset of Agent Skills on GitHub \(arXiv:2608.10906\)](#)
- [GitSkills on Hugging Face \(Parquet mirror\)](#)
- [MSR 2027 Mining Challenge call](#)
- [GitSkills sample repository](#)